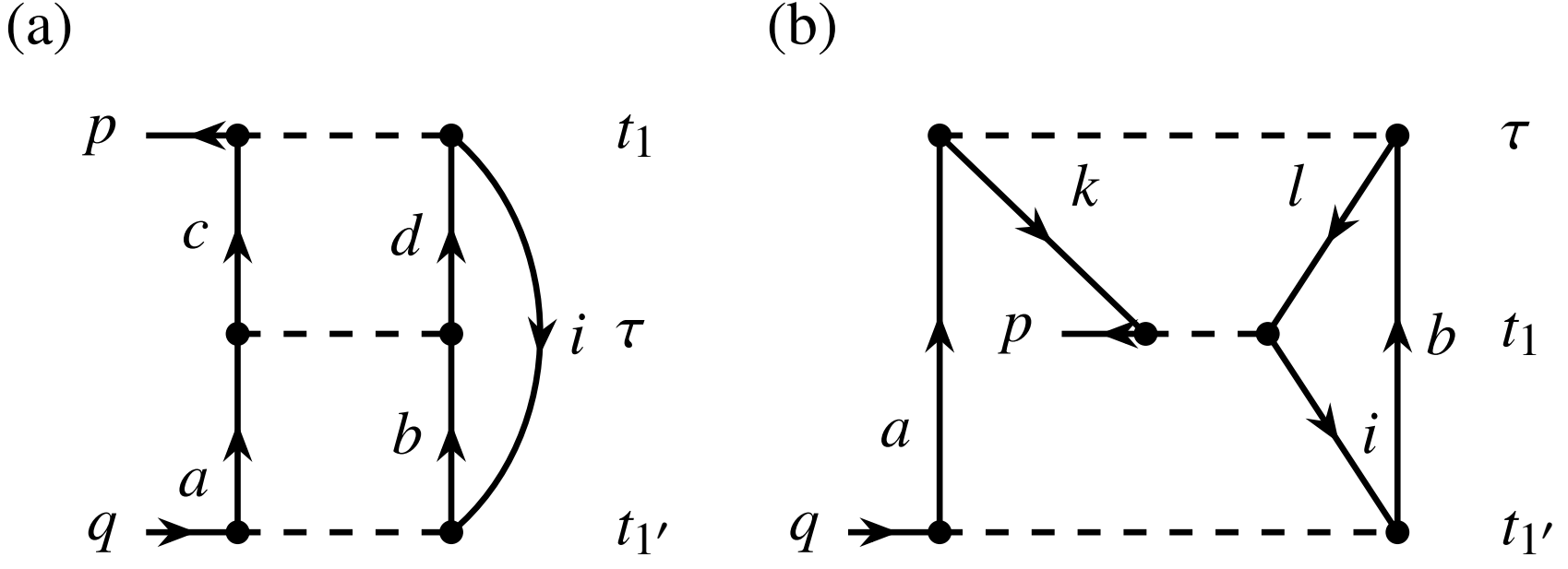

$U^{(1)}\mathcal{G}\,C\,\mathcal{G}\,U^{(1)\dagger}$

$U^{(2)}\mathcal{G}\,U^{(1)\dagger}$